

# AI-BASED PRECISION AGRICULTURE PLATFORM

*Crop Disease Detection • Yield Prediction • Market Intelligence*

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Powered by Computer Vision & Machine Learning

Project Presentation

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# Abstract

Agriculture faces mounting pressure from crop diseases, unpredictable yields, and volatile market prices.

This project proposes an integrated AI-driven precision agriculture platform that leverages computer vision and machine learning to detect crop diseases from leaf images, forecast yield using historical and environmental data, and provide market intelligence for better pricing decisions.

*The system empowers farmers with a single platform that combines diagnosis, forecasting, and decision support — reducing crop loss, improving productivity, and increasing farmer income.*

## 3-in-1

Disease + Yield + Market

## CV/ML

Deep learning powered

## Real-time

Farmer-facing insights

# Problem Statement

1

## Late Disease Detection

Farmers often identify crop diseases only after visible, irreversible damage — leading to major yield loss.

2

## Unreliable Yield Estimates

Traditional yield prediction relies on manual observation and guesswork, with no data-driven forecasting.

3

## Poor Market Visibility

Farmers lack real-time access to market prices and demand trends, resulting in distress selling and low profits.

4

## Fragmented Tools

Existing solutions address only one problem area at a time — no unified platform connects diagnosis, prediction, and pricing.

# Objectives

1

Develop a CV-based model to detect and classify crop diseases from leaf/plant images with high accuracy.

2

Build a machine learning model to predict crop yield using historical, soil, and weather data.

3

Integrate market intelligence to provide farmers with real-time price trends and demand forecasts.

4

Design a unified, farmer-friendly platform accessible via mobile/web for all three services.

5

Enable early intervention and informed decision-making to minimize crop loss and maximize profit.

6

Ensure the system is scalable, low-cost, and usable in low-connectivity rural environments.

# Proposed System

*A unified AI platform that fuses three intelligent engines into one farmer-facing application, connected through a common data and recommendation layer.*

1

## Disease Detection Engine

CNN-based image classifier identifies crop diseases from leaf photos captured via smartphone camera.

2

## Yield Prediction Engine

Regression/ML models forecast expected yield using soil, weather, and historical crop data.

3

## Market Intelligence Engine

Analyzes market price trends and demand signals to recommend optimal selling time and location.

*All three engines feed into a single recommendation dashboard delivered to the farmer.*

# System Flow



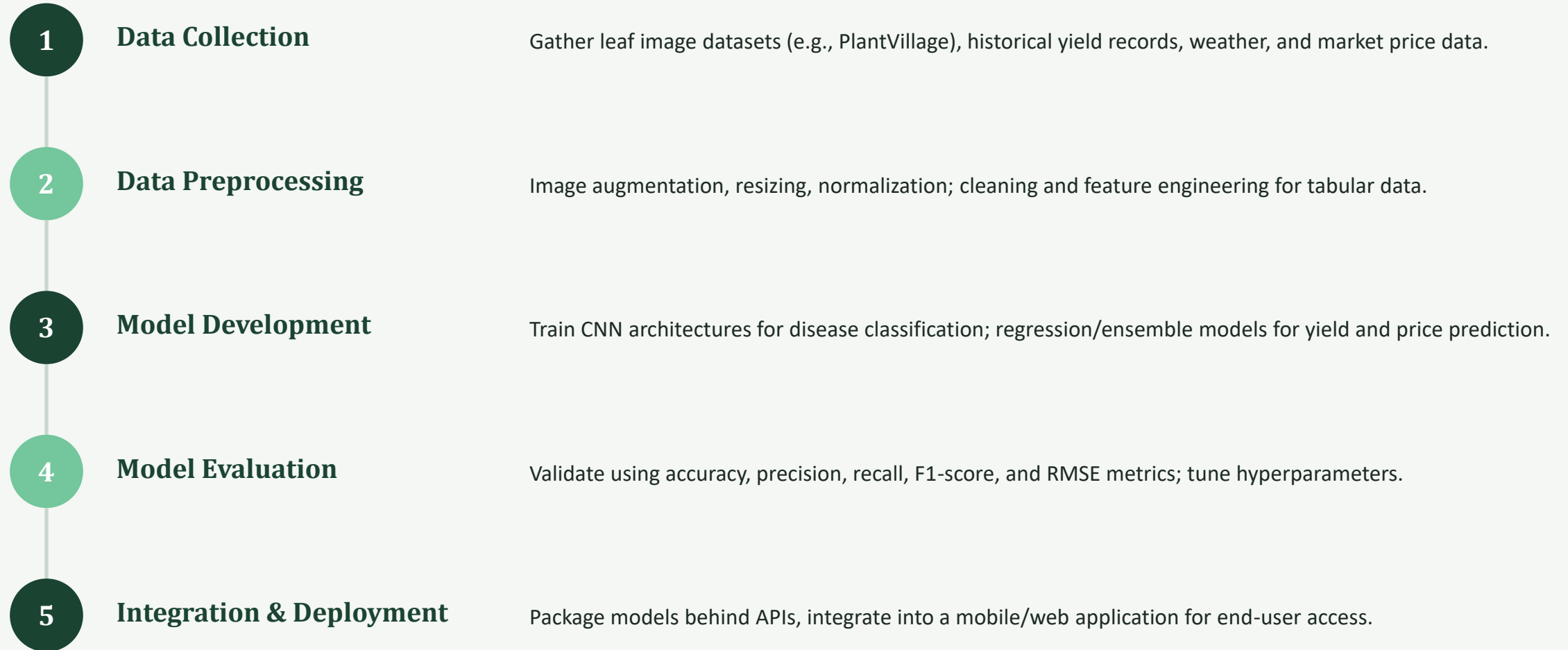
Farmers capture crop images or input field data through a mobile app. The platform preprocesses inputs, runs them through trained CV/ML models, and returns actionable results — disease diagnosis, yield estimate, or price recommendation — on a simple dashboard.

# Diseases Considered

*Representative crop-disease categories targeted by the detection model*

|                                                                                                |                                                                       |                                                                                                   |                                                                                      |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <div>T</div> <div>Tomato</div> <div>Early Blight, Late Blight, Leaf Mold, Bacterial Spot</div> | <div>P</div> <div>Potato</div> <div>Early Blight, Late Blight</div>   | <div>M</div> <div>Maize / Corn</div> <div>Common Rust, Gray Leaf Spot, Northern Leaf Blight</div> | <div>R</div> <div>Rice</div> <div>Bacterial Leaf Blight, Brown Spot, Leaf Smut</div> |
| <div>W</div> <div>Wheat</div> <div>Leaf Rust, Stripe Rust, Powdery Mildew</div>                | <div>G</div> <div>Grape</div> <div>Black Rot, Esca, Leaf Blight</div> | <div>A</div> <div>Apple</div> <div>Apple Scab, Black Rot, Cedar Rust</div>                        | <div>C</div> <div>Cotton</div> <div>Bacterial Blight, Leaf Curl Virus</div>          |

# Methodology





# Technologies Used

## Computer Vision / ML

- TensorFlow / Keras
- PyTorch
- OpenCV
- CNN (ResNet, MobileNet, EfficientNet)
- Scikit-learn

## Backend & APIs

- Python (Flask / FastAPI)
- REST APIs
- Node.js (optional gateway)

## Frontend & Mobile

- React / React Native
- Flutter (alternative)
- Progressive Web App

## Data & Infrastructure

- MySQL / MongoDB
- Cloud Storage (AWS / GCP)
- Docker & CI/CD
- Weather & Market APIs

# System Modules

1

## User / Farmer Module

Registration, profile, farm details, image upload, and query history.

2

## Disease Detection Module

Image classification engine returning disease name, severity, and treatment advice.

3

## Yield Prediction Module

Forecasting engine using soil, weather, and crop history inputs.

4

## Market Intelligence Module

Live price tracking, demand trends, and best-time-to-sell recommendations.

5

## Admin Module

Dataset management, model monitoring, and user/content administration.

6

## Notification Module

Alerts for disease risk, harvest timing, and price fluctuations via SMS/app.

# Expected Output



## Disease Diagnosis Report

Disease name, confidence score, affected area estimate, and recommended treatment.



## Yield Forecast

Predicted yield (quantity per acre/hectare) with confidence range for the current season.



## Market Price Insight

Current and forecasted prices, best nearby markets, and optimal selling window.



## Unified Farmer Dashboard

Single interface consolidating disease alerts, yield outlook, and price trends.

# Advantages



Early disease detection reduces crop loss and pesticide overuse.



Data-driven yield forecasts support better planning and financing decisions.



Market intelligence helps farmers avoid distress selling and get fair prices.



Single unified platform reduces the need for multiple disconnected tools.



Scalable across crops, regions, and languages for wide accessibility.



Low-cost, smartphone-based access suitable for smallholder farmers.

# Conclusion

The proposed AI-based precision agriculture platform brings together crop disease detection, yield prediction, and market intelligence into one accessible system.

By combining computer vision and machine learning, it equips farmers with timely, accurate, and actionable insights — reducing losses, improving yields, and strengthening income through smarter market decisions.

## Detect

diseases early

## Predict

yield accurately

## Decide

when & where to sell

**Thank You**